

THE FIVE COGNITIVE STAGES OF PHYSICAL AGENTS

From Continuous Physical Transduction to Real-Time Discrete Action Invariant Enforcement

STAGE 1: TRANSDUCTION

Physical Photons → Metric Tensors

- CMOS exposure · IMU sense
- MIPI zero-copy DMA views

STAGE 2: WORLD MODELING

Latent Belief & SE(3) Frame Tree

- Dynamic transforms · Spatial map
- JEPA physics & occlusion tracking

STAGE 3: REASONING & INTENT

Goal Grounding & Expiring Leases

- Edge VLM semantic spatial deliberation
- Coarse 3D target box + lease t_{expire}

STAGE 4: ACTION PLANNING

Generative Action Chunking ($H=16$)

- Diffusion / ACT trajectory synthesis
- C^2 continuous quintic spline ensembling

STAGE 5: REAL-TIME REFLEX

Deterministic Barrier Enforcement

- 1 kHz FreeRTOS QP solver: $h(x) \geq 0$
- Dynamic stopping check · Fallback FSM

THE COGNITIVE PIPELINE PRINCIPLE

High-level cognitive models propose intentions over long horizons; low-level reflex circuits guarantee physical safety over microsecond steps.